Colligative properties of electrolyte

21. (b) *NaCl* gives maximum ion hence it shows lowest freezing point

22 **Explanation:**

Depression in freezing point:

$$\Delta T_f = iK_f m$$

Greater the van't Hoff factor i, greater the depression in freezing point, hence lower the freezing point.

- Glucose → non-electrolyte → $i = 1$
- $KNO_3 \rightarrow K^+ + NO_3^- \rightarrow i = 2$
- $AlCl_3 \rightarrow Al^{3+} + 3Cl^- \rightarrow i = 4$

Thus freezing point decreases in order:

$$AlCl_3 < KNO_3 < \text{Glucose}$$

(a) $AlCl_3 < KNO_3 <$Glucose

23. (b) Lesser the number of particles in solution. Lesser the depression in freezing point, *i.e.* higher the freezing point.

24. (c) $BaCl_2$ gives maximum ion hence it shows maximum depression in freezing point.

25. **Explanation:**

Depression in freezing point:

$$\Delta T_f = iK_f m$$

All are non-electrolytes, so $i = 1$.

All solutions have same molality $0.1m$, therefore all produce the same depression in freezing point.

Hence all have the same freezing point.

Answer: All have the same freezing point

26. (d) We know that lowering of freezing point is a colligative property which is directly proportional to the number of particles formed by one mole of compound therefore 0.1*M* $Al_2(SO_4)_3$ solution will have minimum freezing point.

27. (a) $Al_2(SO_4)_3$ gives maximum ion hence its gives lowest freezing point.

28. (b) Colligative property in decreasing order

$Na_3PO_4 > Na_2SO_4 > NaCl$

$Na_3PO_4 \rightarrow 3Na^+ + PO_4^{3-} = 4$

$Na_2SO_4 \rightarrow 2Na^+ + SO_4^{2-} = 3$

$NaCl \rightarrow Na^+ + Cl^- = 2$

29. (d) $K_4[Fe(CN)_6]$ gives maximum ion. Hence it have lowest vapour pressure.